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Hiring Manager
HaloBraid
Cambridge, MA

Dear Hiring Manager,

I am writing to apply for the Embedded Systems Engineer position at HaloBraid. As a robotics engineer with hands-on experience building STM32 firmware, tuning feedback control systems, and consolidating electronics from prototype boards to production-ready PCBs, I can contribute directly to HaloBraid's transition from 450+ prototype iterations to a market-ready braiding robot.

My strongest alignment with this role is in embedded motor control and communication protocols. For a national robotics competition, I wrote embedded C firmware on an STM32F407 that coordinated CAN-based motor control, BLE communication, and three nested PID loops for omnidirectional navigation, achieving sub-2 mm tracking accuracy. I also designed a custom UART protocol with DMA-driven interrupts for non-blocking 100 Hz sensor updates at under 0.2% error rate. Separately, on a remote-mining crane project, I implemented fuzzy PID closed-loop control for stepper motors and built an ESP32-based Wi-Fi link for low-latency multi-device coordination. These systems required the same real-time, multi-protocol embedded architecture that a precision braiding mechanism demands.

Beyond firmware, I bring hardware integration experience critical for a small team shipping a physical product. I consolidated five standalone circuit boards into a single 4-layer PCB using KiCad, cutting board footprint by 40% and verifying signal integrity on a 100 MHz oscilloscope. In an industrial internship, I designed an STM32-based Modbus RTU gateway for a six-axis robotic arm and proposed an encoder-triggered imaging module that reduced point-cloud data loss by 30%. I also led a five-person cross-functional team at UC Berkeley to deliver a hybrid flying-driving robot prototype in ten weeks, the kind of compressed, high-ownership cycle that defines early-stage hardware development.

While my computer vision experience is indirect, rooted in real-time motion-capture integration and encoder-based imaging rather than camera-based CV pipelines, I am a fast study with strong fundamentals in real-time embedded perception. I would welcome the chance to discuss how my embedded controls and hardware background can accelerate HaloBraid's path to product launch.

Sincerely,
Felix Hu